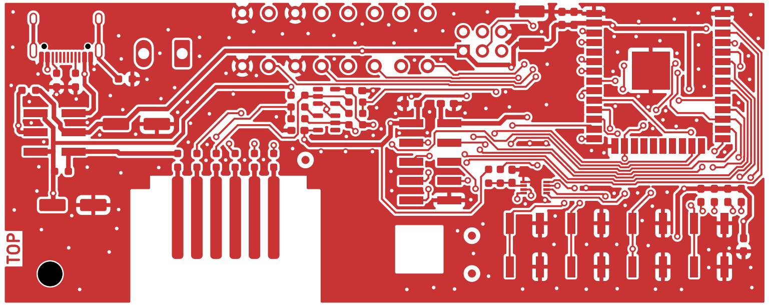
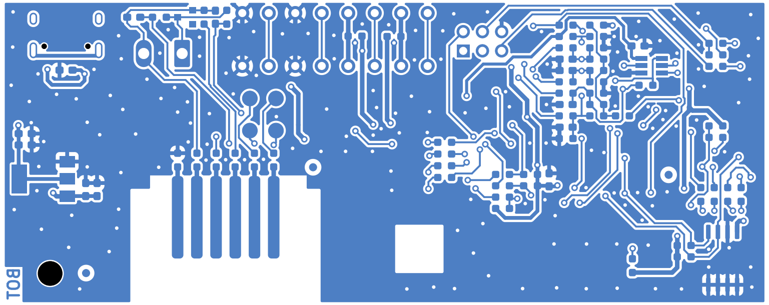
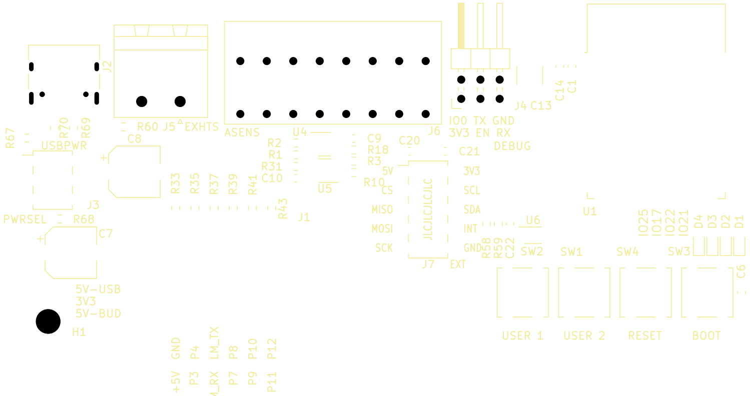
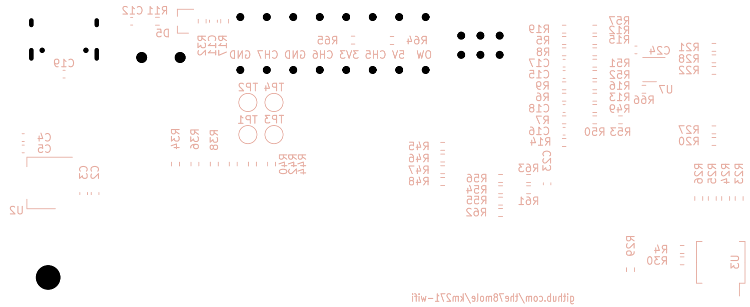
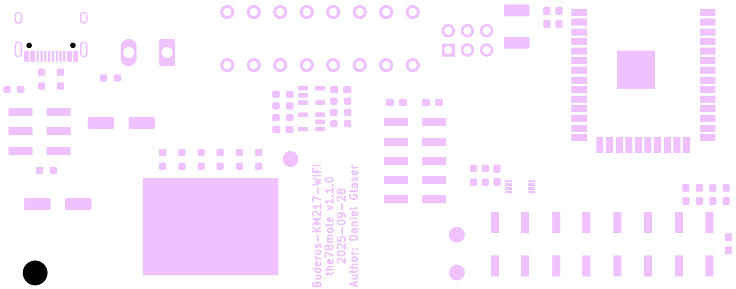
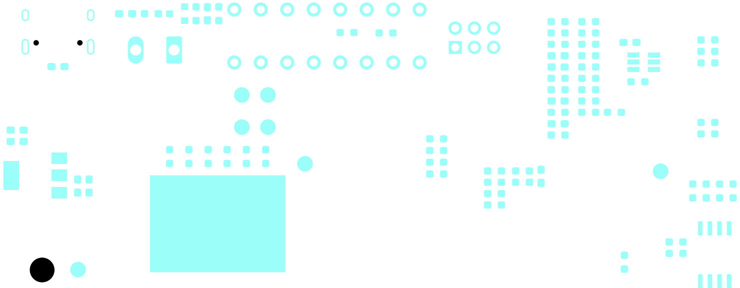

TOP

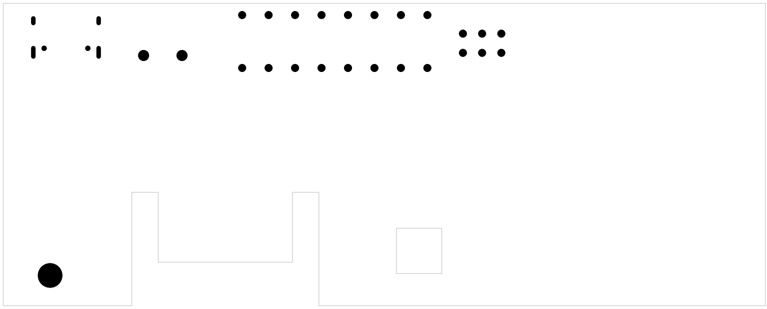

BOT

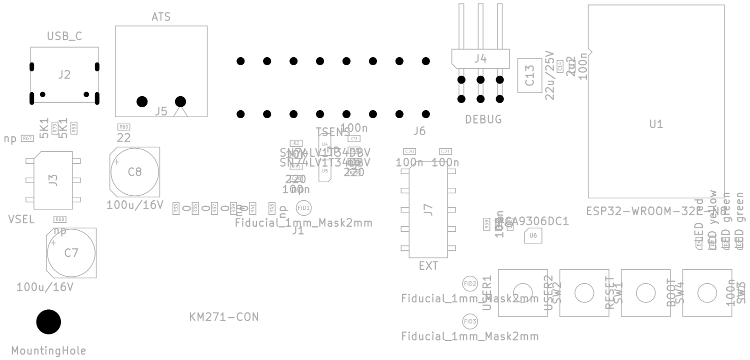

J2
R67
R70
R69
USBPWR
R60
J5
EXHTS
C8
ASENS
U4
U5
R2
R1
R31
C10
R33
R35
R37
R39
R41
R43
J1
C9
R18
R3
R10
C20
J6
5V
CS
MISO
MOSI
SCK
JLCJLCJLCJLC
J7
C21
3V3
SCL
SDA
INT
GND
EXT
IO0
3V3
TX
EN
GND
RX
J4
C13
C14
C1
DEBUG
U1
R58
R59
C22
U6
SW2
SW1
SW4
SW3
IO25
IO17
IO22
IO21
D4
D3
D2
D1
C6
USER 1
USER 2
RESET
BOOT
PWRSEL
R68
J3
C7
5V-USB
3V3
5V-BUD
H1
+5V
GND
P3
P4
LM_RX
LM_TX
P7
P8
P9
P10
P11
P12

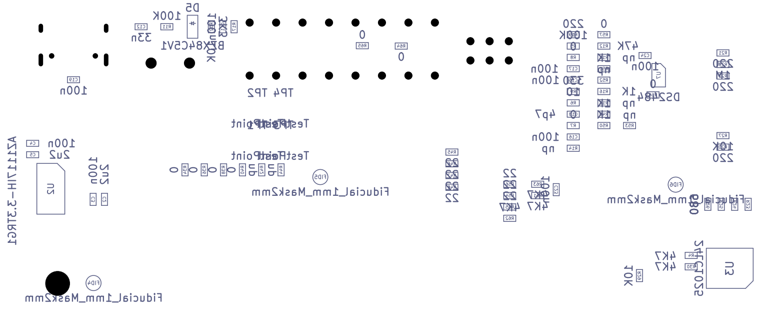

R11 C12
D5
GND CH7 GND CH6 3V3 CH5 5V OW
R65
R64
TP4 TP2
TP3 TP1
R34 R36 R38
C19
C4
C5
C2
C3
U2
R45
R46
R47
R48
R56
R54
R55
R62
R61
R63
R19
R5
R8
C17
C15
R9
R6
C18
R7
C16
R14
R57
R12
R15
R51
R52
R16
R13
R49
R66
U7
C24
R21
R28
R22
R53 R50
R27
R20
R26 R25 R24 R23
R29
R4
R30
U3
github.com/the78mole/km271-wifi

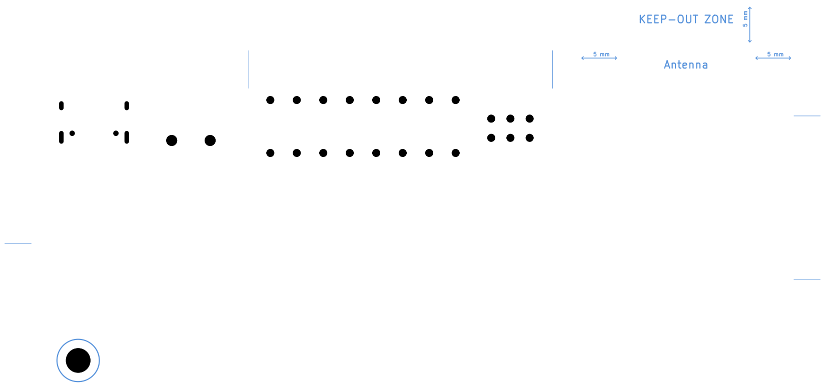

Buderus-KM217-WIFI
the78mole v1.1.0
2025-09-28
Author: Daniel Glaser






USB_C
J2
ATS
J5
J6
J4
DEBUG
C13
22u/25V
2u2
100n
U1
ESP32-WROOM-32E-N8
np
5K1
5K1
22
J3
VSEL
np
C8
100u/16V
C7
100u/16V
TSENS
100n
SN74LV1T34DBV
SN74LV1T34DBV
220
220
100n
np
Fiducial_1mm_Mask2mm
J1
100n
100n
J7
EXT
100n
PCA9306DC1
LED red
LED yellow
LED green
LED green
USER1
USER2
SW2
RESET
SW1
BOOT
SW4
100n
SW3
Fiducial_1mm_Mask2mm
Fiducial_1mm_Mask2mm
KM271-CON
MountingHole


AZ1117IH-3.3TRG1
BZX84C5V1
D5
TP4 TP2
TestPoint
DS2484
Fiducial_1mm_Mask2mm
U2
U3



FAB NOTE J1 (KM271 card edge): 12 gold fingers F+B, 1.6 mm FR4 (+-10%).
Order option: Gold Fingers = YES (ENIG / hard gold on fingers), mask open over fingers.
Bevel: NOT POSSIBLE – finger edge (y=124.25) is recessed 5.75 mm between the board legs
(y=130) and JLC chamfers only boards >= 50x50 mm. Original module is used unbevelled.
Fingers end 0.2 mm before the board edge; no vias/pads within 1 mm (rule area GOLD_FINGERS).

FAB NOTES Buderus-KM217-WIFI rev v1.1.0
Stackup: 2 layers, FR4 TG135, 1.6 mm +-10 pct, outer copper 1 oz (35 um)
L1 F.Cu signal/GND pour | core 1.5 mm | L2 B.Cu signal/GND pour
Finish: ENIG (gold fingers). Gold fingers: YES, bevel: NO (see J1 note)
Mask green, silk white, vias tented. Min track/space 0.20/0.20 mm, min via 0.3/0.6 mm
Assembly: Standard PCBA, both sides. BOM/CPL: production/bom.csv, production/positions.csv